

UMEÅ UNIVERSITY

Co-Creating With AI:

Non-Linear Design Exploration of Biodata-Driven Applications

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Abstract

This thesis investigates how health science experts and non-experts co-create biodata-driven applications with LLM-based code-generation tools. Informed by posthuman and more-than-human perspectives, the study examines co-creation as an entangled process involving participants, prompts, AI systems, biodata, generated interfaces, and situated interpretation. Six participants (two health-science experts and four non-experts) took part in individual co-creation sessions using two AI-supported conditions: a locally running exploratory prototype designed for non-linear design exploration, and the commercial version of Claude. Empirical material consisted of think-aloud verbalizations, post-session interviews, participant prompts, AI-generated artefacts, screenshots, and researcher notes, allowing the analysis to attend to both participants' reflections and the interactional traces of AI-supported design work. The material was analysed using reflexive thematic analysis. The findings show that participants negotiated four central dimensions of co-creation: becoming understandable to the AI, working with the right amount of friction, bargaining authorship, and seeking explanations of AI-generated outputs. The thesis argues that AI programming tools for biodata should support staged, explainable, and non-linear exploration rather than only rapid generation, especially when bodily data becomes material for design decisions and interpretation.

Keywords: Human–AI Co-Creation, AI Programming Tools, Vibe Coding, Biodata, More-than-Human Design, Entanglement

1. Introduction

The development of artificial intelligence has increasingly changed how people engage with programming, design, and digital prototyping. In recent years, AI programming tools have made it possible to generate code-based applications through natural language instructions, reducing the need for users to manually write and debug code. In this mode of interaction, programming becomes less a purely technical activity and more a process of articulation, interpretation, and negotiation between human and AI. The human is no longer positioned as the sole creator of the application, but as someone who guides, questions, selects, and redirects what the system produces (Tholander and Jonsson 2026).

This development is particularly relevant for Human–Computer Interaction (HCI), where design and prototyping have long been understood as exploratory practices. Sketching, low-fidelity prototyping, and iterative design allow ideas to remain open while designers test possibilities, compare alternatives, and gradually refine a direction. AI programming tools extend this logic by turning natural language prompts into executable prototypes (Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026). Within seconds, a prompt can become an interface, a visualisation, or an interactive system. This creates new opportunities for rapid design exploration, particularly for those with domain expertise but limited programming experience. At the same time, however, it alters the conditions of early design

work. AI-generated prototypes can appear polished and complete very quickly, which may narrow the design space before users have had the opportunity to fully explore the problem, question their assumptions, and clarify what they want to create (Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026). In HCI, these interactions are increasingly understood through human–AI co-creation. Rather than treating AI systems as passive tools, this perspective examines how humans and AI systems jointly shape creative and design processes. From posthuman and more-than-human perspectives, agency is not located only in the human. Instead, agency emerges through relations among people, technologies, data, infrastructures, prompts, interfaces, and contexts (Forlano 2017; Frauenberger 2020; Giaccardi and Redström 2020; Nicenboim et al. 2023).

This is important because AI systems do not simply execute human intention. They propose structures, select visual forms, introduce metaphors, generate alternatives, and sometimes move the design in unexpected directions. Co-creation with AI, therefore, involves both possibilities and friction. The system can support ideation by generating options quickly, but it can also disrupt the process by assuming too much, misunderstanding the prompt, or producing outputs that are difficult to interpret (Jonsson and Tholander 2022; Tholander and Jonsson 2026).

These issues become especially significant when AI programming tools are used with biodata. Biodata refers to data about bodies, behaviours, and potentially thoughts and feelings (Sanches et al., 2022). It can include information such as sleep, stress, pain, movement, heart rate variability, emotion, weather, and activity patterns. Although such data often appears numerical and objective, its meaning is situated. Biodata is shaped by how it is collected, processed, displayed, and interpreted. When biodata is used as material for AI-generated applications, the system not only creates an interface, but it also participates in shaping how bodily states, health, recovery, and well-being are represented.

This creates a gap in current research. Existing HCI work has examined AI-supported ideation, co-coding, and design exploration, and related work has also investigated biodata, well-being technologies, and machine learning models of the body (Benjamin et al. 2021; Jonsson and Tholander 2022; Sanches et al. 2022; Tholander and Jonsson 2023, 2026; Zhu et al. 2025; Kaklopoulou et al. 2026). However, less is known about how people use LLM-based code-generation tools to create biodata-driven applications, especially in contexts where users bring health-related or design-relevant expertise but may not be trained programmers. This is important because AI programming tools may make application creation more accessible, but they also require users to translate situated knowledge into prompts that the system can interpret. In domains such as physiotherapy and health-oriented design, this translation is not neutral. It involves clinical reasoning, embodied knowledge, privacy concerns, and careful interpretation of bodily data.

This thesis aims to investigate how participants co-create biodata-driven applications with AI programming tools, focusing on the process of co-creation rather than on the technical quality or functional completeness of the generated applications. The study examines how health science experts and non-experts move from initial ideas to generated prototypes by formulating prompts, interpreting AI-generated outputs, responding to unexpected design directions, negotiating control, and reflecting on the role of AI in shaping biodata-driven

prototypes. To explore these dynamics, the study uses two AI-supported conditions: a locally running exploratory prototype designed to support non-linear design exploration, and the default commercial version of Claude (Anthropic 2026) as a more conventional AI programming tool. Through individual co-creation sessions, think-aloud data, generated artefacts, prompts, screenshots, and interviews, the study addresses the following research question: How do health science experts and non-experts use LLM-based code-generation tools to create biodata-driven applications, and how do they experience the process of co-creating such applications with AI?

2. Background and related work

2.1 AI Programming tools as natural-language sketching

The technological development of Artificial Intelligence (AI) has accelerated significantly in recent years, particularly through the emergence of AI programming tools that enable the generation of code-based applications using natural language. This shift has transformed how individuals engage with programming, moving from a practice centered on manually writing and debugging complex code to one in which users interact with an intelligent system capable of proposing ideas, suggesting alternative solutions, and generating executable code. This phenomenon has recently been described as vibe coding, referring to an approach in which users disengage from low-level technical details and instead focus on conceptual formulation and ideation. In this context, the *vibe* refers to the user articulating intentions, ideas, and directions, while the system interprets and executes these inputs into code (Meske et al. 2025; Sarkar and Drosos 2025). Although the term itself was coined in early 2025, the underlying practices emerged earlier, with AI programming tools becoming publicly available in late 2023 and more rudimentary models appearing from around 2020 (Ray 2025).

In this mode of interaction, attention shifts away from implementation toward the evaluation, interpretation, and synthesis of system outputs. The human is therefore repositioned, not as the sole creator of code, but as an agent who guides, refines, and makes sense of the outputs produced by the system (Tholander and Jonsson 2026). The introduction of these technologies thus reconfigures how application development and code-based knowledge are understood, moving away from a model in which the human is the sole creator toward one in which the human engages as a collaborator with the system. This shift provides a foundation for understanding such interactions as forms of human–AI co-creation.

Within HCI and design research, ideation and sketching have long been considered central practices for exploring possible design spaces. Sketching is commonly conceptualized as a reflective practice through which ideas are externalized and iteratively explored. Drawing on Buxton (2011) and Schön (1992), sketching can be understood as a dialogue between the designer and the emerging representation, where seeing and acting continuously inform each other, enabling the discovery, critique, and refinement of design possibilities. In parallel, low-fidelity prototyping extends this exploratory approach into early

representations of interaction. Typically implemented through simple and inexpensive formats such as paper prototypes, these artifacts support rapid iteration and early evaluation, making it possible to explore multiple design alternatives before committing to a specific solution (Buxton 2011).

Programming in HCI has increasingly been understood as a sketching practice, where code is used to explore ideas rather than implement predefined solutions. Practices such as tinkering, bricolage, and sketching with code emphasize rapid, iterative experimentation, allowing designers to navigate a design space through provisional artifacts (Bergstrom and Blackwell 2016). This perspective is reinforced by work on digital materials, which shows how quick, exploratory prototypes can expose the properties of technologies and open new design possibilities. From this view, computation is not a fixed substrate but a material that can be worked with, probed, and reshaped through design activity (Sundström et al. 2011). Across this literature, programming shifts from a linear engineering activity to a non-linear, material, and exploratory process, aligning closely with sketching practices in design.

This trajectory also extends to AI programming tools, where vibe coding introduces sketch-like qualities into computational prototyping through natural language interaction. Prompts allow participants to externalize early ideas, while generated code and interfaces can function as provisional, executable representations that are quickly explored, evaluated, and revised. The process is therefore iterative and dialogical: designers propose directions, inspect what the system produces, and refine or redirect the output through continued interaction. However, vibe coding should not be understood as a replacement for sketching. Sketching is a broader design practice that includes drawing, annotating, thinking-through-making, negotiating visual and material space, and producing situated or collective forms of knowledge. Sketching can operate as a relational and collective making practice, where knowledge emerges through layers, annotations, intervention, and shared authorship (Vasconcelos et al. 2024). Vibe coding, thus, shares some characteristics with sketching, particularly provisionality, iteration, and generative exploration, but it only extends certain aspects of sketching into a computational medium rather than replacing sketching as a design practice.

At the same time, AI programming tools change the conditions of early design work. Traditional sketching often preserves ambiguity and incompleteness, allowing ideas to remain open while designers explore possible directions. By contrast, AI programming tools can quickly produce outputs that appear complete, polished, or high-fidelity. This can support rapid exploration, but it can also change the rhythm of ideation by moving quickly from an initial formulation to a concrete application structure. Prior research on design ideation with generative AI suggests that such systems may reshape ideation by narrowing the design space at stages where ambiguity is typically productive (Jonsson and Tholander 2022; Tholander and Jonsson 2026). Consequently, the relevance of AI programming tools for design lies not only in their ability to accelerate prototyping, but also in how they alter the relationship between early ideas, generated artifacts, and design exploration.

2.2 Human-AI co-creation and design exploration

The development of this phenomenon has been anticipated by earlier work within the Human-Computer Interaction (HCI) community, which has extensively examined how human and non-human actors engage with and co-constitute one another. While a substantial part of the field has traditionally been grounded in human-centered design, in which design materials are shaped primarily to serve human needs and position humans as the central agents, more recent perspectives have moved beyond this view. These approaches advocate for a broader understanding of design that operates at a more-than-human level, in which humans and non-human entities are considered interdependent and mutually constitutive. Within this perspective, agency is not exclusively located in the human but is distributed across the relations between humans, technologies, and other material actors, shifting the focus from human dominance to relational co-construction (Redström 2006; Baumer and Brubaker 2017; Forlano 2017; Frauenberger 2020; Giaccardi and Redström 2020; Nicenboim et al. 2023).

Within this context, Frauenberger (2020) introduces the notion of entanglement as a key theoretical perspective in HCI. Entanglement conceptualizes humans and technological systems not as separate entities that interact, but as fundamentally interconnected and co-constituted through their relations. Accordingly, agency is not located solely in the human; rather, it emerges through ongoing intra-actions between human and non-human actors, where boundaries and roles are continuously enacted rather than predefined. In a similar vein, Forlano (2017) frames posthumanism as a theoretical position that challenges the traditional human-centered view of the world by questioning the idea of the human as a rational, autonomous, and central agent. Instead, it emphasizes the relational and interconnected nature of humans, technologies, and environments, highlighting how agency, knowledge, and experience emerge through socio-technical and ecological assemblages. Within this context, AI systems can be seen as a concrete site where these theoretical positions become visible in practice, as they make explicit the distributed and relational nature of agency in design processes.

Research in HCI has begun to examine how designers co-create with AI programming tools, focusing on both their opportunities and limitations, as well as the characteristics of these systems as design materials and how they entangle with human practices (Benjamin et al. 2021; Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026; Zhu et al. 2025; Kaklopoulou et al. 2026). In particular, machine learning introduces inherent uncertainty, as outputs are probabilistic, variable, and partially opaque rather than fixed or fully predictable (Benjamin et al. 2021). This reframes sketching with AI as an engagement with uncertain materials, where generated artifacts are provisional and require interpretation. Consequently, design emerges through an ongoing negotiation between human intent and system-generated variation, positioning co-creation as a process in which both human and AI actively shape the exploration of the design space.

Within this emerging body of work, human–AI interaction is often conceptualized as a form of asymmetrical co-creation. AI systems contribute by generating variations, alternatives, and suggestions, while the human remains responsible for interpreting, selecting, and situating these outputs within a broader design trajectory (Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026). Across these studies, the interaction

is described as one in which AI supports breadth, speed, and variation, while the human provides contextual judgment and direction. This dynamic is important for AI-supported design because generated outputs do not simply execute predefined ideas. They can also influence what users notice, what they compare, and what directions they consider possible.

This does not mean that AI systems understand the design context in the same way as human collaborators. Prior work emphasizes that interaction with generative systems is shaped by the user's need to formulate, adjust, and reinterpret prompts and outputs. Natural language prompting operates as a form of programming through articulation and refinement. Users formulate prompts not only to influence system responses, but also to clarify their own design intentions. This positions interaction as both a means of control and a reflective practice, where design direction develops through iterative engagement with the system (Tholander and Jonsson 2026).

This body of work is relevant to the present study because it frames AI programming tools not simply as technical instruments, but as systems that can shape design exploration. The focus is therefore not only whether such tools generate functional applications, but how users work with them while moving from an initial idea to a generated prototype. This is particularly important in contexts where users may have domain expertise but not necessarily formal training in programming or interaction design. In such cases, AI programming tools may become a way of making application ideas more tangible, while also requiring users to translate professional or experiential knowledge into forms that the system can process.

2.3 Biodata as a Situated Material for AI-Supported Application Creation

While co-creation with AI programming tools has been explored within HCI, less attention has been given to how these tools shape the creation of biodata-driven applications. This is important because biodata is not a neutral input for design. Within HCI, biodata refers to "data about people's bodies, behaviors, and more controversially, their thoughts and feelings" (Sanches et al., 2022, p. 2). Rather than being a neutral representation, it is co-produced through interactions between bodies, sensors, and computational processes. Sanches et al. (2022) argue that biodata emerges from "entangled factors including people, context, and the particular sensors, analysis, and display techniques employed" (Sanches et al. 2022, 1), embedding design decisions into what becomes visible.

Building on this, Merrill, Chuang and Cheshire (2019) show that biodata is not passively read but actively interpreted in everyday contexts, where users construct meaning based on personal assumptions, social situations, and prior experiences. As a result, interaction with biodata is not about accessing objective information, but about negotiating and making sense of situated and often ambiguous representations. This interpretive quality becomes especially important when biodata includes information related to sleep, stress, pain, movement, emotion, weather, heart rate variability, and other bodily or contextual states. Such data may appear numerical or objective, but its meaning depends on how it is collected, modeled, displayed, and used. This understanding of biodata becomes more complex when

biodata is integrated into machine learning (ML) systems. In ML, biodata is translated into computational models that classify, predict, and personalize aspects of human bodies and behavior, requiring complex and situated experiences to be reduced into discrete variables and patterns (Zhu et al. 2025). However, this process embeds specific assumptions about the human body, often privileging regularity, stability, and predefined categories (Zhu et al. 2025). As a result, ML does not simply represent biodata. It reconfigures biodata into a computational construct shaped by design and modeling decisions, which can create tensions with lived bodily experience.

This integration of biodata into ML systems becomes particularly salient in the context of vibe coding, where designers directly engage with data through AI-mediated sketching practices. Recent work shows that LLMs enable designers to rapidly prototype and explore biodata-driven applications, effectively turning data, models, and code into materials for iterative sense-making and design exploration (Kaklopoulou et al. 2026). In this setting, human–AI co-creation is defined through a continuous interaction in which designers articulate ideas through prompts while the system generates executable representations that shape the design trajectory. Kaklopoulou et al. (2026) identifies three key characteristics of this interaction. First, LLMs force the early articulation of vague or embodied phenomena, requiring designers to translate complex bodily experiences into explicit textual descriptions, which can constrain exploration (Kaklopoulou et al. 2026). Second, the interaction tends to lead to local optima, where the system reinforces familiar patterns and makes it difficult to reframe or diverge from initial directions (Kaklopoulou et al. 2026). Third, generated prototypes exhibit what is described as *soufflé fidelity*, appearing complete and functional while remaining fragile and difficult to modify (Kaklopoulou et al. 2026). Together, these findings show that co-creation with AI in the context of biodata is not a neutral or supportive process, but one that actively shapes how data is conceptualized, represented, and explored. Vibe coding enables rapid and high-fidelity engagement with biodata, but at the same time introduces constraints related to articulation, exploration, and material stability. As a result, design emerges through an ongoing negotiation between human interpretation of biodata and the generative, yet structuring, capabilities of AI systems.

Building on this literature, the present study investigates how health science experts and participants with HCI/AI backgrounds use LLM-based code-generation tools to create biodata-driven applications. While AI programming tools have been explored in clinical contexts for data analysis, education, and rehabilitation (M. H. Lee et al. 2021; Chow and Ng 2025; Y. Lee and Huh 2025; Pesce and Cheungpasitporn 2025; Ong et al. 2026), and separately, co-creation with AI programming tools and biodata exploration has been studied (Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026; Kaklopoulou et al. 2026). Less is known about how these areas converge when biodata becomes the material for AI-supported application creation. This gap is particularly relevant in expert-oriented contexts such as physiotherapy, where bodily data, professional judgment, and situated interpretation are closely connected. Studying this process is necessary because LLM-based tools do not only help users generate applications; they also shape how ideas are formulated, how data is represented, and how possible applications become imaginable. The present study therefore explores how participants move from initial ideas to generated prototypes,

how they experience the process of co-creating with AI, and how LLM-based tools may support or constrain the exploration of biodata-driven design possibilities.

3. Methodology

3.1 Research design

This study adopted a qualitative research approach to examine how participants collaborated with AI programming tools to co-create biodata-driven application concepts. The study did not aim to evaluate the technical performance, usability or functional completeness of the generated applications. Rather, it focused on how participants explored design possibilities, articulated prompts, interpreted AI-generated outputs, negotiated control with the system and reflected on the role of AI during the creative process. While the study was informed by co-design and participatory design traditions, it did not take the form of a collective co-design workshop. Instead, individual co-creation sessions were used to examine how participants interacted with two AI programming tools while generating application concepts from biodata. This one-to-one format was chosen because the study aimed to capture participants' situated reasoning, uncertainty, and interpretation during interaction with the systems. The focus was therefore on the process of co-creation rather than on the production of a finalized or validated design outcome.

This framing follows HCI work that treats generative AI systems not simply as passive instruments, but as systems that can shape ideation, creative direction, and users' expectations during design work (Jonsson and Tholander 2022; Tholander and Jonsson 2026). In this study, AI programming tools were approached as co-creative systems that participate in the design process by producing suggestions, interface structures, visual metaphors, and working prototypes from natural language prompts. This perspective also aligns with more-than-human and posthuman approaches in HCI, which question the idea that human actors are the only source of agency in design processes. Rather than treating the AI as a neutral tool, the study examined how agency, authorship, and meaning were distributed across participants, prompts, generated outputs, biodata, interface structures, and the wider study setting.

The study used two contrasting conditions. These conditions were not designed as experimentally equivalent systems. Instead, they were used as two different interactional configurations through which participants could reflect on different forms of AI-supported co-creation. The first condition used a locally running exploratory prototype designed to support non-linear design exploration. The second condition used the default commercial version of Claude, developed by Anthropic AI (Anthropic 2026), representing a more conventional AI programming tool based on natural language prompting and iterative refinement.

3.2 Study Context and Systems

The first condition involved a prototype developed by Associate Professor Pedro Sanches. The prototype was designed to support open-ended exploration of the design space while keeping biodata locally on the computer. This local configuration was important because the broader aim of the prototype was to investigate how domain experts, particularly physiotherapists and health-oriented practitioners, might use biodata to create applications relevant to their professional practice. By avoiding the upload of personal or sensitive data to an external server, the prototype supported a more privacy-preserving and GDPR-sensitive workflow. The prototype was built as a wrapper around Anthropic's Claude model Opus 4.7 (Anthropic 2026). The system used an exploration-tree structure, where users could begin from one idea and then create multiple branches from it. This structure was intended to support a slower and more exploratory design process than commercial AI programming tools, which often move quickly from a prompt to a complete high-fidelity application.

The prototype offered several ways of developing an application concept. First, participants could use a classic prompt-based mode, in which they described in natural language what they wanted the AI to generate. From each generated block, participants could continue the same design direction, create further branches, or move into alternative trajectories. The system also included an "Explore variations" function, which generated three alternative versions from the current idea. A clarification function allowed participants to ask or receive clarification before continuing. A "Keep/change" function allowed participants to preserve selected elements of a generated prototype while modifying others. Finally, a "What else?" function prompted the system to move in a different direction and generate an alternative outside the current design trajectory. The branching option of the system 1 is shown in Figure 1, while the rest of the functions can be found in Appendix 1. The functionality of System 2 can be found in Figure 2.

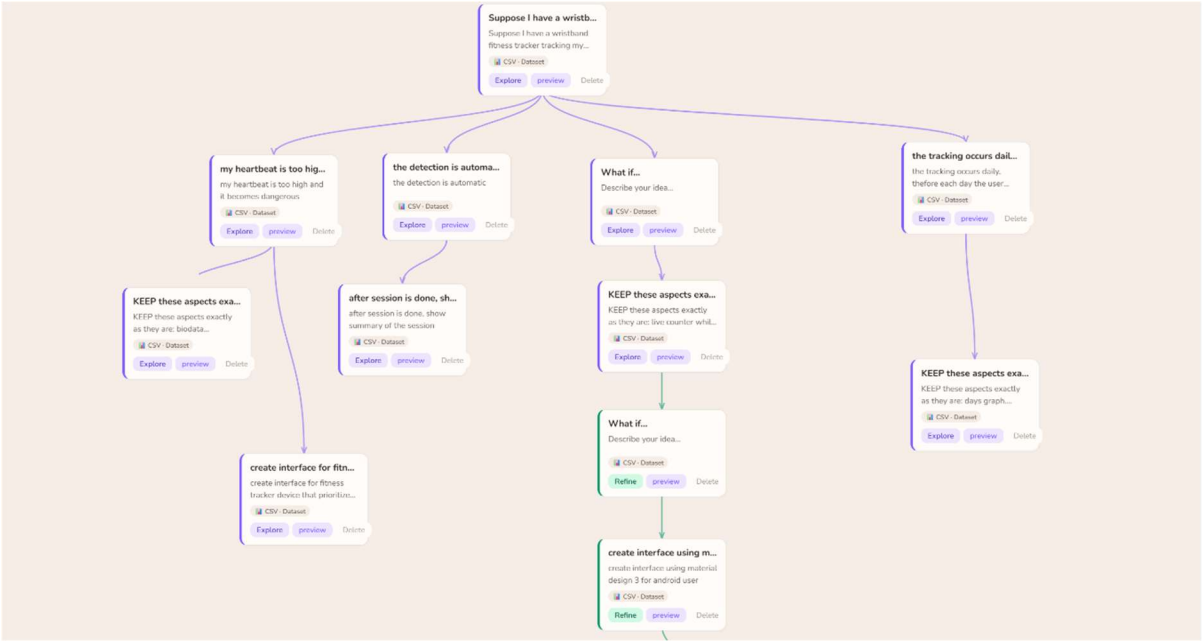


Figure 1. System 1's branching feature, which allows multiple design directions to be developed from a single block.

The second condition involved the default commercial version of Claude Opus 4.7 (Anthropic 2026). In this condition, participants interacted directly with Claude through natural language prompting and refined the generated application through subsequent instructions. This condition represented a commercially available form of AI programming or “vibe coding,” where users describe an application idea and receive a generated prototype or code-based output. Compared with the exploratory prototype, this system produced more polished outputs but offered less explicit support for branching, parallel exploration, and selective comparison between alternative design paths. Because the commercial system may involve sending data to external servers, participants did not use personal or sensitive biodata in this condition. Instead, synthetic datasets were created for the study. The use of synthetic data allowed participants to work with plausible biodata scenarios while avoiding unnecessary privacy risks. For one participant, with a background in physiotherapy and research, the dataset was tailored around exercises and movement-related data relevant to Anterior Cruciate Ligament (ACL) rehabilitation. This choice was made because the participant's research field is digitalization, physiotherapy, and ACL rehabilitation; tailoring the data to her area of competence was intended to make her more comfortable designing with biodata. For participants with HCI, AI, design, or broader health-technology backgrounds, the dataset was adapted toward a more general wearable-device scenario,

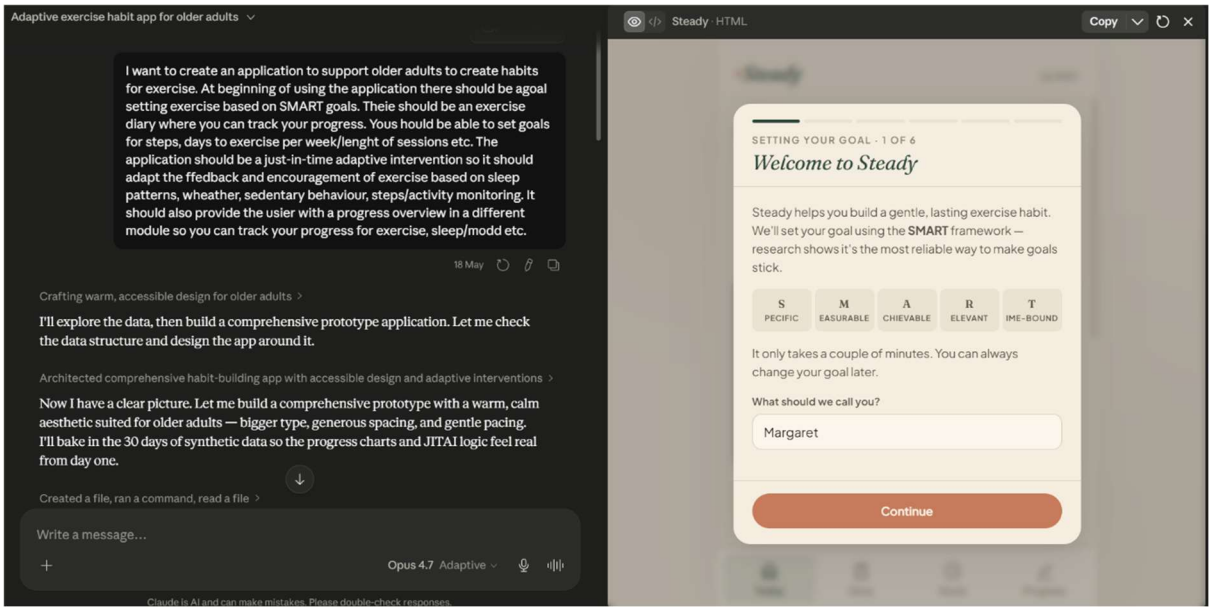


Figure 2. The functionality of System 2. The prompt is text-based, and the LLM responds with a text-based description of what it is doing. The final interactive prototype is shown on the right.

including biodata that could plausibly be collected through consumer tracking technologies. These included sleep score, sleep duration, heart rate variability, resting heart rate, stress score, pain level, step count, emotion, weather, and temperature. More details regarding the datasets can be found in APPENDIX 3.

3.3 Participants

The study involved six participants recruited through Umeå University and professional networks connected to HCI, AI, health science, and design-oriented technology use. Participants were selected because they could provide relevant perspectives on AI-supported application creation, biodata interpretation, or design exploration. In Table 1, you can find participant identifiers, their profession and background, and the design intentions during the co-creation session.

ID	Profession/background	Design intentions
P1	PhD in physiotherapy	Designing a system that could help interpret the data
P2	Associate professor in physiotherapy and digital health	Create an adaptive exercise-support app for older adults
P3	Student with a background in CS and AI	Design a smartwatch UI that could detect stress from biodata
P4	Student with a background in CS and HCI	An app that could help with PCOS (Polycystic Ovary Syndrome)
P5	Student with a background in UX and HCI	A training schedule app that correlates health with biodata
P6	Student with a background in front-end and HCI	Training app that focuses on heart-rate detection and weather

Table 1. The table represents the participants' identifiers, their profession and background, and the design intention during the individual co-creation session

3.4 Data collection techniques

Each participant took part in an individual co-creation session lasting approximately one to two hours. The sessions followed the same general structure. First, participants were introduced to the study, the concept of biodata, the synthetic dataset, and the two AI systems. The researcher explained that the purpose was not to create a perfect application, but to understand how participants used the systems to explore, interpret, and develop application ideas from biodata. Participants then completed the first condition using the

local exploratory prototype. After this, they completed the second condition using the default version of Claude. The order was fixed across participants, with the local prototype always used first. This order was chosen because the prototype required more explanation and because the study aimed to examine participants' experiences of the exploratory system before they interacted with the more familiar commercial interface.

During both conditions, participants were asked to think aloud while creating an application from the provided dataset. The think-aloud procedure was used to capture participants' in-the-moment reasoning, uncertainties, interpretations, and reactions while working with the systems. Participants were encouraged to verbalize what they were trying to achieve, how they interpreted the AI's responses, what they found useful or problematic, and how they decided what to keep, change, reject, or develop further. The think-aloud material complemented the retrospective interview by showing how participants made sense of AI outputs during the co-creation process.

Participants were not given a fixed application template to reproduce. Instead, they were free to decide what kind of application they wanted to create from the dataset. Some participants approached the task from a professional or clinical perspective, while others explored more personal, speculative, or design-oriented applications. This openness was important because the study focused on ideation, interpretation, and co-creative negotiation rather than task completion. The researcher interacted with participants during the sessions to explain available functions, answer procedural questions, and prompt reflection when relevant. This interaction was treated as part of the study setting rather than as a source of contamination to be removed. The sessions, therefore, produced data not only about individual participants' isolated thoughts, but about how participants made sense of the tools within a situated interaction involving the researcher, the system, and the dataset.

After completing both conditions, participants took part in a short semi-structured interview; the interview questions can be found in Appendix 2. The interview focused on their experience of working with each system, how they perceived the differences between the two conditions, how they understood the role of AI in the design process, and whether the tools supported or constrained their ability to explore ideas. The interview also invited participants to reflect on how such tools might be used in professional or design-oriented contexts.

The collected material included audio recordings, think-aloud verbalizations, semi-structured interview responses, participant prompts, generated outputs, application artefacts, screenshots, and researcher notes. These materials were treated together as evidence of the co-creation process. The analysis, thus, considered not only what participants said, but also what they wrote, what the AI generated, how the artefacts changed, and how participants responded to those changes.

3.5 Data Analysis

The collected material was analysed using reflexive thematic analysis (RTA) (Braun and Clarke 2021; Byrne 2022; Rigby and Iacovides 2026). RTA was chosen because the study aimed to interpret patterns of meaning across participants' experiences of AI-supported co-creation, rather than quantify usability outcomes, measure task success, or evaluate the

technical quality of the generated applications. This approach was appropriate because the study focused on meaning-making, uncertainty, control, authorship, explanation, and the situated interpretation of biodata-driven design outputs. The study adopted an interpretivist qualitative orientation. Participant accounts were not treated as transparent reports of an objective experience, but as situated accounts produced through interaction with the systems, the dataset, the researcher, and the study setting. In line with RTA analysis, themes were understood as analytic constructions developed through active engagement with the data, rather than as categories that simply emerged from the material. This distinction is important because RTA treats researcher subjectivity as a resource for analysis, not as a problem to be eliminated.

The analysis began with repeated familiarization with the dataset. This involved listening to audio recordings, reading transcripts, reviewing participant prompts, examining generated artefacts, and revisiting researcher notes. During this phase, attention was given to how participants moved between intention, prompting, AI output, interpretation, critique, and iteration. The artefacts were not analysed separately from the transcripts. Instead, they were examined in relation to the participant's stated intention and subsequent response.

The primary analytic unit was the co-creative episode. A co-creative episode was understood as a sequence involving a participant's intention, a prompt or interaction, an AI-generated output, and the participant's interpretation, rejection, acceptance, modification, or questioning of that output. Because the data involved interaction between participants, prompts, AI-generated outputs, biodata, and design decisions, analysing isolated statements alone risked overlooking how meaning, intention, and authorship developed across the interaction. The co-creative episode, developed for this study, was therefore used to guide attention to these sequences without imposing a rigid deductive coding frame. This allowed the analysis to remain open to participants' accounts while also capturing the entangled structure of AI-supported co-creation.

Initial coding was conducted manually by the first author. Coding began close to the explicit content of participants' statements and actions, focusing on how participants described their intentions, articulated prompts, interpreted AI-generated outputs, experienced control or constraint, and responded to unexpected system behaviour. Codes were examined in relation to the study's broader concern with co-creation, biodata, agency, uncertainty, and more-than-human design. Analytic memos were written throughout the analysis. These memos documented interpretive decisions, recurring tensions, uncertainties, and possible relationships between codes.

Candidate themes were developed through recursive movement between coded extracts, participant prompts, generated artefacts, and analytic memos. The analysis followed the six recursive phases commonly associated with reflexive thematic analysis: familiarization with the data, coding, generating initial themes, developing and reviewing themes, refining and naming themes, and writing up the final analytic account (Braun and Clarke 2021; Byrne 2022). These phases were treated as an iterative process in which earlier interpretations were revisited and revised as the analysis developed. Discussions with the thesis supervisor were used as reflexive dialogue rather than as a procedure for achieving coding agreement.

The final themes were developed to explain how participants experienced AI-supported co-creation, how the two tools shaped their exploration of biodata-driven application ideas, and how AI programming tools may support or limit open-ended design thinking in expert and non-expert contexts. The themes are analytic constructions that tell a coherent story about how participants negotiated intelligibility, friction, authorship, and explanation when working with AI-generated biodata applications.

3.6 Ethics Considerations

The study involved biodata-driven application creation, but participants did not use sensitive personal biodata in the commercial AI condition. Synthetic datasets were created to avoid unnecessary privacy risks while still allowing participants to work with realistic and meaningful biodata scenarios. In the local prototype condition, the system was configured so that uploaded data remained on the local computer. This supported a more privacy-preserving workflow and allowed the study to examine how AI programming tools might be used with biodata in ways that are more sensitive to GDPR and data-protection concerns. Participants were informed about the purpose of the study, the use of audio recording, the collection of prompts and generated artefacts, and their right to withdraw. They were also informed that screenshots or records of their generated applications could be used as part of the analysis. No video recording was used. All participant names were replaced with identifiers, and identifying details were removed or minimized in the reporting. All storage and data processing were done in the local Umea University servers.

Because biodata can easily be interpreted as intimate or health-related, the analysis treated participants' responses to generated outputs with particular care. The study did not evaluate participants' health, bodies, or personal well-being. Instead, it examined how participants interpreted and negotiated AI-generated representations of biodata.

4. Results and Data Analysis

This chapter presents four themes developed through the analysis. The first, becoming understandable to the AI, concerns how participants worked to make their intentions legible to the system. The second, the right amount of friction, concerns when interruption in the design process became productive rather than obstructive. The third, bargaining authorship, concerns how control over the design was distributed between participant and system. The fourth, seeking explanation, concerns participants' need to understand why the AI produced what it did. The two systems are treated not as tools to be ranked, but as contrasting design conditions that shaped how participants explored, questioned, and redirected AI-generated prototypes. Each theme is illustrated with participant quotes, prompts, and generated outputs from the co-creation sessions.

4.1 Can you Understand Me?

This theme captures how participants experienced prompting as a negotiation of intelligibility. They were not only formulating design ideas, but also trying to infer what kind of language, specificity, and domain knowledge the AI system could understand. After the participants had been introduced to the co-creation sessions, the prompting activity revealed a recurring uncertainty around how to become understandable to the AI. This uncertainty was not only technical. Participants were not simply learning how to write better prompts; they were trying to estimate the linguistic and conceptual boundaries of the system. They had to decide whether to use professional terminology, ordinary language, detailed requirements, direct commands, or exploratory questions. This made the beginning of the design process fragile, because participants had to think not only about what they wanted to create, but also about how the AI might interpret their intention.

P2, a physiotherapist specialized in digital health, entered the session with a clear direction, prompting: "I want to create an application to support older adults to create habits for exercise. At the beginning of using the application, there should be a goal-setting exercise based on SMART goals. There should be an exercise diary where you can track your progress. You should be able to set goals for steps, days to exercise per week/length of sessions, etc." (P2). However, after generating the first prototype, she reflected that "I was not sure how I should articulate it" (P2). She was also surprised that the AI understood what SMART goals were and could translate them into the application structure. This shows that even when participants had a clear design idea, they were still uncertain about how to address the AI and how much they needed to explain.

P6 described a similar experience, stating: "I need to remember that I am speaking to an AI, not to a person" (P6). When asked why, he answered: "I need to use a specific sentence structure. I should not use double negatives. I need to be very straightforward and give it commands and orders instead of having a dialogue as if I were speaking to a person. I need to think more about what I want in order to formulate it" (P6). For P6, the interaction required a shift from ordinary dialogue to a more controlled and command-based form of communication. He argued that failing to do this could lead to a poor outcome or to a model that did not understand the task.

This is significant because the participants were working in an early design process, where the purpose was to explore a design space. In this context, worrying about how to formulate ideas because of fear of being misunderstood can slow down the creative process. It can also limit ideation to what participants believe the AI is able to understand, rather than what they actually want to explore. P1 expressed this concern by asking how to "know what language this system is built on and if I can use physiotherapist wording or if I need to put it in a different way, how technical you can be" (P1). This suggests that the participants were not only designing with the AI; they were also trying to infer the system's boundaries.

P3 felt that System 1 did not always understand his intention. He stated: "It left me wondering whether it was just contextualising the prompt based on keywords, and not really paying attention to every word I wrote" (P3). Here, the issue was not only whether the system produced a good or bad result, but whether it engaged with the meaning of the prompt. P3 perceived the system as extracting keywords and producing an output based on them, rather than interpreting the instruction as a whole. He also added: "It did not ask me for

clarification" (P3). This points to an important limitation in the interaction: understanding and communication were experienced as one-sided.

From a more-than-human perspective, this suggests that sense-making between human and non-human actors cannot be reduced to passive execution. If the AI system is treated as a co-creative participant, then the interaction needs to support the negotiation of a shared direction. The problem is not simply that users need to become better at prompting. Rather, the interaction itself needs to support mutual adjustment between the participant and the system.

This theme shows that co-creation with AI-programming tools begins before the first prototype is generated. It begins in the participant's attempt to become legible to the system. In this study, participants often adjusted their language, narrowed their ideas, or became more command-based because they were unsure what the AI could understand. This matters because early design work depends on ambiguity, partial ideas, and situated knowledge. If participants feel that only certain kinds of language can be understood by the AI, the design space may already be narrowed before generation begins.

4.2 The Right Amount of Friction

Friction had different functionalities during the co-creation session; notably, friction became valuable when it interrupted the design process in ways participants could still interpret, redirect, or use as design material. The difference between the two systems made this tension visible. System 1 was designed to keep the design process open through branching, variation, and non-linear exploration. Instead of moving directly from a prompt to a finished prototype, it allowed participants to build gradually, compare alternatives, and follow different paths. This was valued by P3, who criticized how AI programming tools often generate too much too quickly: "Sometimes you want to start with something basic when designing, but you give a prompt and it builds the whole complete thing with mock data and things you do not need. It blows up the prototype, while you wanted to start from something simple and build from there" (P3). For P3, the problem was not only the quality of the output, but the way the system collapsed early ideation into a finished-looking product before the user had shaped the foundations of the idea.

In this sense, friction supported agency when it slowed down the design process and kept the participant involved. P3 described the value of moving "from one box to another," where one box could act as a scaffold and later boxes could build on it through different paths. This suggests that participants did not necessarily want AI systems to remove uncertainty. They wanted uncertainty to be structured in a way that supported exploration. The tree-like structure of System 1 therefore offered a different rhythm of co-creation: one in which the participant could move through partial possibilities rather than accept or reject a single complete application.

However, this openness also made the system's instability more visible. Participants were sometimes confronted with features or metaphors they had not asked for, and these moments did not always become productive. P2 reacted negatively when the system added an audio feature to an exercise-support application: "I feel skeptical because I do not know what it is. Maybe it is some kind of stress-reducing feature, but I do not know. It is irritating,

like when ChatGPT starts doing things you never asked for. I feel a bit annoyed" (P2). The issue here was not only that the system diverged from the prompt, but that the divergence was not understandable as a useful design move. The output created friction, but not reflection.

A similar issue appeared in P4's reaction to the system's repeated use of garden metaphors in a prototype related to PCOS, meal tracking, workout habits, and stress. P4 rejected this framing strongly: "Why is everything a garden? No. It feels like talking about a girl coming of age, like the girl has bloomed. No. We are not a garden. This is very biased. It is cringe behaviour" (P4). Here, friction was not only aesthetic or functional. It touched the participant's relationship to biodata, gendered assumptions, and bodily experience. The metaphor imposed a symbolic interpretation of the body that felt inappropriate and reductive. This shows that, in biodata-driven design, unexpected outputs carry particular weight. They do not only represent interface choices; they can also frame how bodies, emotions, health, and wellbeing are understood.

P6's response to an overloaded fitness-tracker prototype also shows how friction can become obstructive. The participant asked for an interface that prioritized important information, but the system produced too many elements: "But this is not what I wanted at all. I said a fitness-tracker device that prioritizes important information, but this is way too much stuff. Recovery. Also, this thing is not clear at all" (P6). In this case, the system contradicted the participant's core intention without making that contradiction useful. Rather than opening the design space, the output created confusion and extra work.

At the same time, friction could become productive when the output gave participants something they could take up and transform. P3, for example, did not want to use the whole prototype but identified a specific visual idea: "I liked the circle filling up with a full emoji. I could take that idea and integrate it into something else" (P3). The value of the AI output was therefore not in the complete result, but in a fragment that could be extracted and reworked. Similarly, P5's interaction with the "Body Weather" concept began with confusion but became generative. P5 first wondered whether the system meant literal weather, then reinterpreted it as a metaphor for bodily states: "Maybe it could be like, 'Your body is in a rainy mood right now; stay inside.' Or maybe, 'It is bright and sunny; get out there'" (P5). In this case, the system's unexpected direction opened a new metaphorical space connecting biodata, mood, weather, and activity.

The comparison with System 2 clarified this balance. While System 1 sometimes created too much instability, System 2 often reduced friction by producing a more complete application from the first prompt. P5 described this as a "bigger leap": "It took a bigger step and said: Okay, here is a big app" (P5). This was useful when participants wanted a polished result, but it also risked closing the design space too early. The interaction became oriented toward refining an already-formed prototype rather than exploring multiple possible directions.

This theme contributes the idea that friction in AI-supported co-creation is not inherently good or bad. Its value depends on whether the participant can do something with it. System 1 created more opportunities for surprise, branching, and reinterpretation, but these same qualities also made the system feel unstable when outputs became arbitrary, overloaded, or

insensitive to the meaning of biodata. System 2 reduced some of this friction by producing more complete outputs, but this also risked closing the design space too early. The productive zone was therefore between randomness and over-resolution: participants needed enough interruption to rethink the design, but enough structure to remain critically involved.

4.3 Bargaining Authorship

Authorship was not experienced as belonging either to the human or to the AI, but as something bargained through the interactional structure of the tool. The two systems distributed authorship differently. System 2, as a conventional AI-programming tool, relied mainly on natural language prompting. Participants described what they wanted, and the system generated a relatively complete, high-fidelity application. This could be useful when the participant had a clear idea and wanted a polished result. However, when prompts were short or not detailed, the system often filled in the missing parts by itself. It assumed intentions, selected visual structures, added features, and produced an app that looked almost finished. System 1, by contrast, kept authorship more open through branching, variation, deletion, and re-generation. Its outputs were less polished, but the process allowed participants to explore several directions without committing too early.

This difference was especially important in relation to critical reflection. P1 described how System 2 could take too much initiative: “It takes a lot for granted. And that might not be so good, because we are lazy. Maybe we just accept it and think, okay, this looks great, and then we go for it instead of really asking ourselves what we want to have. So this might be a limitation for critical analysis. It might create a lack of critical reasoning, and that is a big part” (P1). This quote shows that authorship was not only about creativity, but also about responsibility. In health-related contexts, participants saw it as important that clinicians remain active in deciding what the system should do and why. If the AI makes too many choices too quickly, the expert risks becoming someone who approves or corrects an already-made output, rather than someone who shapes the design from within.

P6 raised a similar concern when reflecting on how the system generated too many graphs and explanations. The issue was not only that the output contained too much information, but that the system had decided which representations were relevant without asking: “I scrolled down and saw an explanation of the pattern, and I thought, okay, very good. But then there was another graph, and another one, and another one. It was too much. Maybe it would be better if I could decide which type of graph I am interested in” (P6). Here, authorship concerns the right to select, prioritize, and frame the data. This is especially important in biodata-driven applications, where visualizations do not simply display information. They shape how bodies, stress, recovery, movement, pain, and progress are interpreted.

Several participants imagined a better AI-programming tool as one that would support a slower and more staged process. P6 connected this directly to design practice: “It should be more reflective of how designers work. First we do sketches, then low-fidelity prototypes, and then high-fidelity prototypes, color contrast, design systems, and so on. It should help me through the different phases of my work instead of just giving me a finished output immediately. I want to create it from scratch with the system, slowly” (P6). This does not

suggest that participants wanted less AI involvement. Rather, they wanted AI involvement to happen at the right level of fidelity and at the right moment. The system should support the movement from vague intention, to sketch, to prototype, to refinement, instead of immediately deciding the whole application.

P3 expressed this desired relation through the metaphor of building a house: “I would have something very basic at the top, almost a plain canvas with just the basic scaffold. Then I would build from there. It is like having an empty room where the sockets and the door are already there, but you can paint it whatever colour you want and change the roof. It is like building a house. You start with the foundation and then slowly build. Then you decide what to have inside each room. Of course, I want to be the architect and have agency” (P3). This metaphor summarizes the participants’ desired form of authorship. They did not want to build alone, but they also did not want the AI to furnish the whole house before they had decided what kind of space they were making. The AI could provide scaffolding, structure, alternatives, and material, while the participant remained responsible for the direction.

System 1 was valuable because it made this negotiation more visible. Its branching structure allowed participants to move in several directions, compare alternatives, and return to previous ideas. P6 described this flexibility clearly: “I can go multiple ways at the same time without always having to wait for one thing and then correct it. In the prototype, I can add another variation and start a new branch” (P6). P6 later added: “I value flexibility, not just a vertical system. I want a system where I can diverge in different directions, go back, and be more flexible. It can also be a bit more chaotic” (P6). This suggests that authorship may require some degree of productive disorder. A linear system can generate coherent outputs, but it may reduce the user’s role to prompting and correction. A branching system may be less polished, but it preserves the user’s ability to compare, abandon, redirect, and decide.

This theme contributes a specific account of authorship as interactional rather than individual. In these sessions, authorship was shaped by how quickly the system moved, how complete the output appeared, and how easily participants could branch, revise, reject, or return to earlier ideas. System 2 sometimes positioned participants as evaluators of an already-formed output, while System 1 allowed authorship to remain more open through partial prototypes and parallel paths. The central issue was therefore not whether AI should contribute to the design, but how users could decide which parts of the process to delegate and which parts they needed to keep authoring themselves.

4.4 I want to know why

This theme shows that participants needed explanations not only to trust the system, but to keep participating critically in the co-creation process. The contrast between the two systems made this need visible. System 2 often provided text-based explanations alongside the prototype, while System 1 generated interfaces with little or no account of what had happened. For some participants, this made System 1 harder to interpret. P6 described this directly: “I want the system to explain what happens and why” (P6). The issue was not only that the prototype was incomplete or visually confusing, but that the participant could not

access the reasoning behind it. Without explanation, the AI's output appeared as a finished surface detached from the process that produced it.

For P1, explanation was especially important in relation to biodata. The participant wanted to understand "what it is built on" and how different variables were connected. This included not only the calculations behind the prototype, but also the relationship between variables such as pain, performance, and other physiological measures. P1 explained that it would be useful "to see connections also between variables" and to choose specific variables in order to understand "how that affects" the outcome (P1). Here, explanation was not framed as a generic description of the interface. It was connected to the participant's domain expertise and to the interpretive complexity of biodata. A system that creates applications from bodily data needs to show how it moves from data to design, because the meaning of the output depends on how those relations are constructed.

Other participants wanted explanation at the level of design decisions. P3 noted: "Design-wise, I would like to know why it chose this oval-shaped button and why it used the pulsing effect. I think that would give it more grounding, for example if it explained that it follows some design rule" (P3). This quote shows that explanation was not only about transparency in a technical sense. It was also about grounding the prototype as a design object. When the system produced a button, animation, graph, or layout, participants wanted to know whether this was random, convention-based, data-driven, or guided by some design rationale. In this sense, explanation could help distinguish between arbitrary generation and meaningful design support.

P5 made a similar point, but emphasized that explanations should be lightweight: "Maybe I would have liked a short explanation, like, 'Based on the weather data, we are trying this style.' Something minor" (P5). This suggests that participants did not necessarily want long textual justifications. They wanted timely explanations that helped them understand the system's direction without interrupting the flow of ideation. Explanation therefore needed to be available, but not overwhelming. It should support interpretation at the moment where the output becomes surprising, unclear, or potentially useful.

P6's response to an alternative graph illustrates this need clearly: "I suppose it is some kind of two-axis graph, but I cannot understand it. It is very confusing. I like this kind of thing. It is a bit alternative compared with the usual calendar where you scroll and select the date, but I would also like to understand why the AI did this" (P6). The participant did not reject the design. In fact, they recognized that it could be interesting because it differed from a conventional calendar interface. However, the lack of explanation prevented the participant from evaluating or developing the idea further. This shows that explanation can transform confusion into usable ambiguity. Without explanation, an alternative design may remain opaque; with explanation, it can become material for reflection and iteration.

P6 proposed a more interactional form of explanation through an "explain this" function: "It would be great if there were an explanation button. I could click a section and say, explain this. Then a second window could appear and explain why it created that section, or give references, images, links, or something" (P6). This would make explanation situated rather than general. Instead of receiving a long description of the whole prototype, the user could

ask about a specific component at the moment it becomes relevant. This would also preserve the user's control over when explanation is needed.

This theme contributes the traceability dimension of co-creation. Participants did not only want better outputs; they wanted access to the reasoning, assumptions, and design choices behind those outputs. This was especially important for biodata, where visualizations and recommendations can shape how bodies, health, stress, recovery, and performance are interpreted. Explanation therefore functioned as a way of keeping the prototype open to critique. When participants could ask why, the AI-generated interface became something they could inspect and negotiate. When they could not, the output risked becoming a surface that looked designed but remained difficult to question.

5. Discussion

The purpose of this study was to investigate how people engage in non-linear exploration of biodata through AI-programming tools. More specifically, the study examined how health science experts and non-experts co-create with an AI-programming system, and how ideation, exploration, interpretation of biodata, and end-user application concepts emerge through this process. Inspired by posthuman and more-than-human perspectives, the study conceptualizes co-creation as an entangled process in which agency and meaning are not located in the human participant alone, but emerge through the interaction between human expertise, AI systems, biodata, prompts, generated outputs, and interface structures. In doing so, the study contributes to emerging HCI research on human–AI co-creation and AI-supported design exploration (Jonsson and Tholander 2022; Tholander and Jonsson 2023, 2026), while also extending discussions on how biodata and human bodies are represented, interpreted, and reconfigured through AI-mediated design processes (Sanches et al. 2022; Zhu et al. 2025; Kaklopoulou et al. 2026).

Overall, the findings show that AI-programming tools can support rapid ideation for biodata-driven applications, but only when humans remain able to interpret, question, and redirect the co-creation process. Participants valued how quickly the systems transformed ideas into prototypes, yet they also struggled with making their intentions understandable, managing unexpected outputs, preserving authorship, and understanding why the AI made particular design decisions. The central claim of this discussion is that co-creation with AI-programming tools should not be understood as a prompt-to-output workflow. Rather, it should be understood as a negotiated design process in which intelligibility, friction, authorship, and explanation are shaped by concrete interaction design choices.

The results show that communication between humans and AI systems is not only a matter of better prompting. Participants had to infer what kind of language the system could understand, including whether to use professional terminology, ordinary language, detailed requirements, direct commands, or exploratory questions. This supports recent work on AI co-coding and vibe coding (Jonsson and Tholander 2022; Tholander and Jonsson 2026), where users search for a workable instruction syntax while responding to unexpected system behaviour. It also aligns with Tholander and Jonsson's (2023) discussion of AI-supported design ideation, where natural language interaction shapes users' expectations of what the

system can understand and produce. However, the present study adds that this negotiation becomes more complex when biodata and domain knowledge are involved. Health science experts were not simply describing interface preferences; they were translating embodied, clinical, and situated knowledge into computationally actionable prompts. AI-programming tools, therefore, do not remove expertise from the process, but relocate it into articulation, interpretation, and repair.

The second key finding concerns friction. Across their studies of creative AI-based programming, Jonsson and Tholander (2022, 2026) describe friction as a quality that can both interfere with and support creative work. The present findings partly support this view. Participants sometimes used unexpected outputs as design material, especially when a visual element, metaphor, or alternative branch opened a new direction. However, friction was not automatically useful. It became productive only when participants could understand, appropriate, or redirect it. When the system produced overloaded interfaces, unexplained features, or inappropriate metaphors, friction became obstructive rather than reflective. This distinction is important because biodata is not neutral design material. Generated representations of stress, sleep, pain, exercise, mood, or recovery can shape how bodies and wellbeing are interpreted.

This also connects to Benjamin et al. (2021), who argue that Machine Learning (ML) uncertainty should be treated as a design material rather than only as a problem to be solved. For Benjamin et al. (2021), ML outputs are shaped by data and model uncertainty, making them variable, probabilistic, and partially opaque. The present study extends this argument into AI-programming with biodata. System 1 supported uncertainty through branching, rapid variation, and non-linear exploration, allowing participants to keep multiple design possibilities open. System 2 often reduced uncertainty by producing more polished outputs, but this also risked closing the design space too early. In this sense, both systems exposed different strengths and weaknesses. System 1 supported exploration and visible branching, but sometimes lacked explanation and stability. System 2 supported polished generation and text-based feedback, but sometimes assumed too much and positioned the participant as an evaluator of an already-formed output.

The third key finding concerns explanation. Morreale et al. (2026), in their paper *Emergent, not Immanent: A Baradian Reading of Explainable AI,* challenge the dominant view of XAI as a technical process of revealing meanings that already exist inside the model. They argue that interpretation is not immanent to the model, but emerges as a material-discursive performance through situated entanglements between the AI model, human interpreters, datasets, tools, and contexts (Morreale et al. 2026). This perspective helps clarify why participants in the present study did not simply ask the system to reveal its internal logic. They wanted to understand why a particular graph, metaphor, feature, or relation between variables had become meaningful within a specific co-creative situation. Explanation was therefore not detached from the design process. It was part of the interaction between participant, AI system, biodata, prompt, interface, and generated prototype.

From this *Baradian* perspective, the ambiguity observed in the results should not be understood only as system failure. Morreale, Serrà and Mitsufuji (2026) argue that different

entanglements can generate different explanations, and that multiple or even contrasting explanations should be staged and negotiated rather than reduced to one authoritative account. In this study, some ambiguous AI outputs were useful because they invited reinterpretation, while others were rejected because participants could not see how they related to the data or design intention. This suggests that the design challenge is not simply to remove ambiguity, but to create interactional resources through which users can compare outputs, explanations, and design directions. Meaning emerges through comparison, rejection, repair, and reinterpretation.

This is especially important for biodata. Sanches et al. (2022) define biodata as data about bodies, behaviours, and sometimes thoughts and feelings, and argue that biodata is produced through entanglements between people, contexts, sensors, analysis, and display techniques. Merrill, Chuang and Cheshire (2019) similarly show that people interpret biosensor data through existing beliefs about the body and what sensors can reveal. Zhu et al. (2025), further argue that ML systems can embed assumptions about bodies, health, regularity, and normality. In the present study, this became visible when participants resisted metaphors or representations that did not fit their professional or lived understanding of bodily experience. The generated prototype did not simply display biodata; it participated in producing a possible interpretation of the body.

The fourth key finding concerns authorship. Participants did not want the AI to automate the entire design process, but they also did not want to work without AI support. Authorship was continuously bargained through prompts, outputs, corrections, deletions, re-generations, and decisions about what to keep or abandon. This supports more-than-human and posthuman HCI perspectives, where agency is not located solely in the human but distributed across human and non-human actors (Frauenberger, 2020; Giaccardi and Redström, 2020). The present study makes this theoretical position concrete by showing that authorship is enacted through interface design. The speed of generation, the fidelity of the output, the possibility to branch, and the availability of explanation all shaped how much control participants felt they had.

5.1 Limitations

This study has several limitations. First, the participant group was highly diverse, spanning health science experts alongside people from HCI, AI, and design backgrounds. While this mix yielded broad insight into different co-creation styles, it means no definitive claims can be made about any single professional group. A more targeted sample, such as one focused exclusively on physiotherapists, would have allowed a deeper account of how specific clinical expertise shapes the interpretation of biodata and AI tools.

Second, the two systems were not experimentally equivalent. System 1 was an early, exploratory prototype, whereas System 2 was a mature, commercial tool. Some reactions to System 1 therefore likely stemmed from basic usability glitches or system instability rather than from the core concepts of branching and non-linear exploration. The study was intended throughout as an interpretive examination of two different interaction setups, not as a head-to-head experimental test of tool performance.

Finally, the dataset differed for one participant compared with the other five. While this could have changed the results of that specific co-creation session, the aim of the study was not to evaluate the content or the outputs of the sessions themselves, but to understand how participants explored the design possibilities, interacted with the two different LLMs, and perceived the AI-generated outputs. The specific type of biodata is therefore of secondary importance. The study nonetheless acknowledges that, from a methodological standpoint, it would have been stronger had every participant received a distinct synthetic dataset, which would have made this choice more coherent.

Furthermore, the study relies on synthetic datasets. While necessary for privacy and ethical reasons, this means the data may lack the emotional, clinical, or practical weight of real patient tracking. Combined with the short, exploratory nature of the sessions, the findings capture immediate, instinctive reactions rather than how these tools would fit into long-term clinical or design workflows.

5.2 Design implications

The findings suggest that future AI-programming tools should treat understanding, uncertainty, authorship, and explanation as interactional qualities rather than secondary features. First, tools should support staged fidelity. Users should be able to choose whether they want brainstorming, clarification, low-fidelity sketching, visual variation, or high-fidelity implementation. This would reduce the risk of premature closure, where the system generates a polished application before the user has explored the design space. This implication aligns with Buxton's (2011) and Schön's (1992) view of sketching as a reflective and provisional practice, and with Kaklopoulou et al.'s (2026) concern that AI-generated prototypes can appear complete while remaining fragile and difficult to modify.

Second, tools should support non-linear exploration through branching, return, comparison, and merging. Participants valued being able to move in different directions, but future tools should also allow users to combine useful fragments from different branches. Third, explanation should be situated and optional. Instead of only providing long general explanations, systems could allow users to ask "explain this" about a specific graph, visual metaphor, feature, or recommendation.

The findings also suggest that AI-programming tools for biodata should support diffractive forms of explanation. Following Morreale et al. (2026), this means moving away from one fixed explanation and toward interfaces that stage multiple situated interpretations. For example, a user could ask why a graph was created, why a metaphor was selected, or why certain variables were connected, and then compare alternative explanations produced under different prompts, branches, datasets, or visualizations. This would make explanation part of the co-creation process itself, and it would also foreground ambiguity, disagreement, and uncertainty as materials for reflection rather than problems to be hidden.

Fourth, interaction should not rely only on text. Especially in health science and physiotherapy, knowledge is often embodied, visual, and situated. Future systems could support sketches, diagrams, movement examples, annotated datasets, voice input, or visual references as part of the co-creation process. Finally, tools for biodata should support

privacy-sensitive workflows, including local processing where possible, because biodata is intimate and often health-related.

5.3 Future work

Future work should examine AI-programming tools with more targeted participant groups, separating studies with health science experts from studies with non-experts or HCI practitioners. This would make it possible to understand how different forms of expertise shape prompting, interpretation, and authorship. Future studies should also explore the use of richer biodata, including not only physiological data but also emotions, moods, environmental factors such as weather, and situated contextual information. This would support a more complex understanding of the intersection between humans, biodata, AI systems, and non-human environmental actors.

Further research should examine longer-term use in real or semi-real professional contexts. This would show whether non-linear exploration, staged fidelity, and situated explanation remain useful beyond short experimental sessions. Future work could also use speculative design and design futures to explore the ethical and epistemic dilemmas raised by AI-mediated biodata design. Such methods could help researchers and participants imagine alternative forms of human–AI co-creation where meaning is not predefined by the system, but emerges through ongoing interaction between human and non-human actors.

Taken together, the findings suggest that AI-programming tools for biodata should not only generate applications faster but should support the emergence of understanding through situated interaction, where intelligibility, friction, authorship, and explanation are continuously negotiated between human and non-human actors.

6. Conclusion

This thesis examined how participants co-created biodata-driven application concepts with AI programming tools, focusing on how ideation, interpretation, and design direction were shaped through interaction with the systems. The study shows that AI-supported application creation cannot be understood only as a faster way to produce prototypes. Rather, it is a negotiated process in which participants, prompts, generated interfaces, biodata, and system structures jointly shaped what became possible to imagine, question, and develop.

The findings show that co-creation began before the first output was generated. Participants first had to make themselves understandable to the AI, deciding how much design and domain knowledge, detail, and instruction to provide. This made prompting an act of interpretation, not merely a technical command. The study also shows that friction played an ambivalent role. When friction was understandable and redirectable, it supported exploration, surprise, and reflection. When it was arbitrary, overloaded, or disconnected from the meaning of biodata, it disrupted the process and reduced participants' sense of control. Authorship was similarly distributed and unstable. Participants did not reject AI agency, but they wanted to decide when the system should scaffold, suggest, branch, or produce a more complete prototype. For this reason, explanation became central:

participants needed to know why the system had created specific visualizations, metaphors, recommendations, or interface elements to remain critically involved.

These findings extend existing work on human–AI co-creation into the context of biodata (Kaklopoulou et al. 2026; Jonsson and Tholander 2022; Tholander and Jonsson 2026). The study's contribution is to show that when biodata is the material, the generated prototype does not simply display bodily data; it participates in producing a possible interpretation of the body. Generated metaphors, graphs, and visualizations advance one reading of stress, recovery, pain, or other bodily states over others, which is why intelligibility, friction, authorship, and explanation matter more acutely here than in co-creation more generally.

Overall, the thesis contributes to HCI by showing that AI programming tools for biodata should support staged, explainable, and non-linear exploration rather than only rapid generation. Designing for this kind of co-creation means preserving ambiguity, supporting traceability, and allowing users to bargain agency throughout the process, especially when AI systems translate sensitive bodily data into design decisions during early design work.

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APPENDIX 1

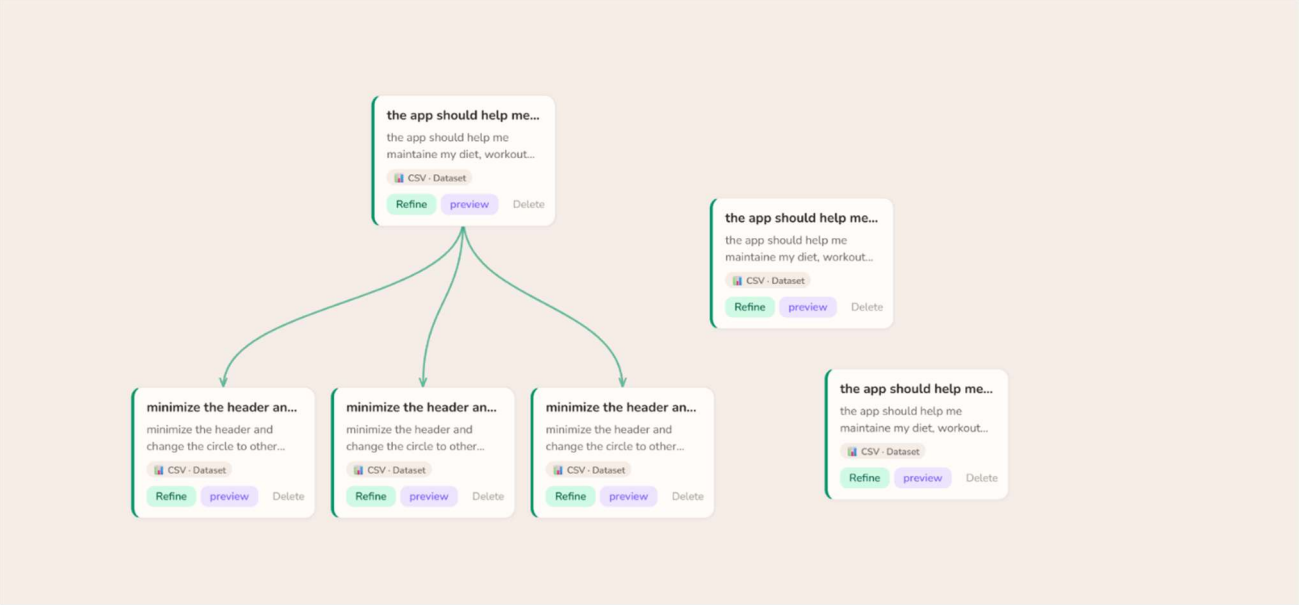


Figure 5. The pictures shows the three variations function. Where three variations of the same prototype are created

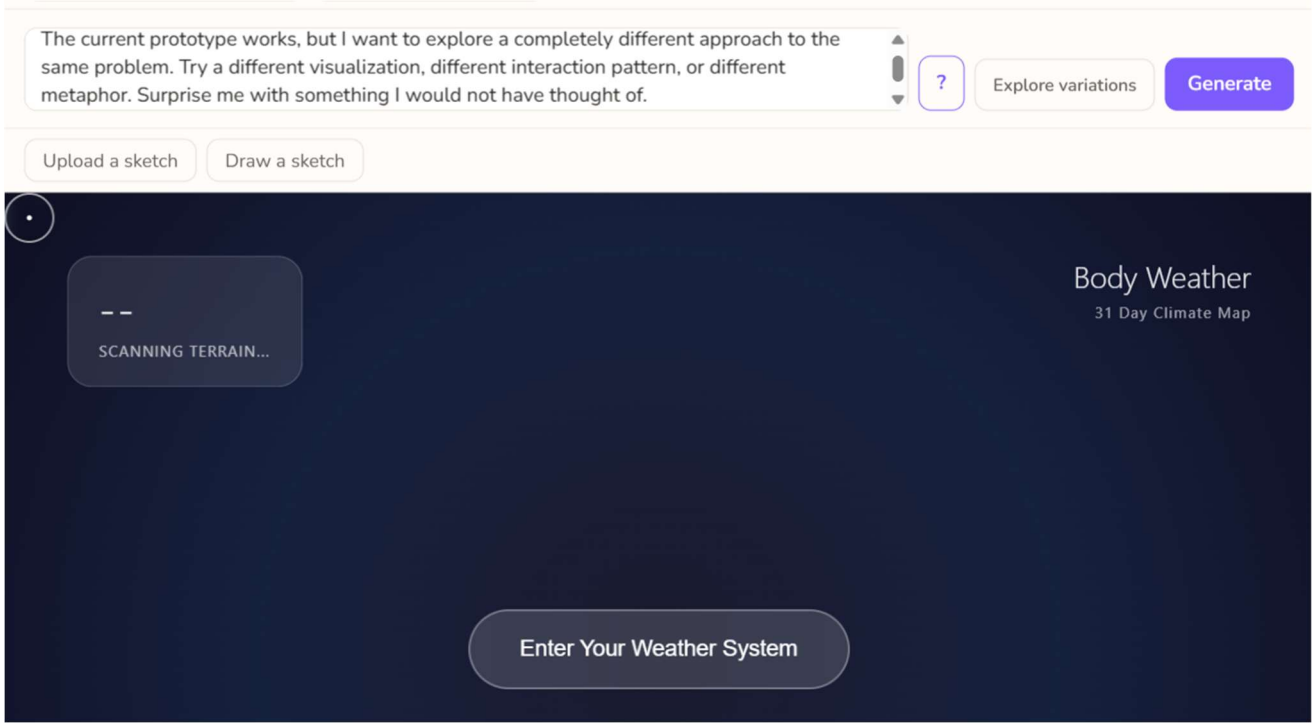


Figure 6. In this pictures, we have illustrated the prompt that will be generated when clicking on the "what else" function

Appendix 2

1) How was your experience designing with these tools?

2) How was your experience using the two different systems?

   - Can you describe any differences you noticed between them?

3) How would you describe your ideation process during the session?

   - How was it to explore ideas with System 1?

   - How was it to explore ideas with System 2?

4) How was your experience working with the two different datasets?

   - Did you notice any differences between them?

5) Did you encounter any limitations during your ideation process?

   - If so, which system influenced your process the most, and in what way?

6) How was your experience communicating with the AI systems?

   - Can you describe any differences between the two versions?

7) Once you began designing, how easy or difficult was it to explore different ideas?

8) If you imagine an ideal system for this type of work, what features would it include?

   - Did any of the tools you used reflect aspects of this ideal?

9) Would you trust the first system by using your own biodata?

Appendix 3

In this appendix are described the two excel files used as biodata for the participants. The first dataset contains a synthetic wearable recovery dataset. It includes 30 daily entries for one fictional participant, labelled P01. Each row represents one day of data and combines physiological, behavioural, subjective, and contextual variables. The dataset includes sleep score, sleep duration, heart rate variability, HRV status, resting heart rate, stress score, perceived energy, pain level, soreness level, step count, active minutes, emotional state, weather condition, temperature, precipitation, recovery status, and short daily notes.

The second file contains a synthetic rehabilitation-oriented movement dataset used in relation to the think-aloud design session. It includes 129 time-series rows from a fictional ACL rehabilitation scenario. The dataset describes three trials of a single-leg drop-landing task performed with the left injured limb. The trial contexts include baseline, after external cue, and late-session fatigue.

Each row represents one time point in milliseconds during the movement sequence. The dataset covers several movement phases, including pre-contact, initial contact, early loading, peak loading, absorption, and stabilization. The recorded variables include vertical ground reaction force normalized to body weight, knee flexion, knee valgus, hip flexion, trunk flexion, ankle dorsiflexion, knee flexion moment, frontal knee displacement, centre-of-pressure movement, quadriceps EMG, hamstring EMG, gluteus medius EMG, pain rating, and apprehension rating.